Xiao Tan

Undergraduate Research Assistant, Hardware Security Lab @ UNC, Chapel Hill, North Carolina, United States
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RESEARCH INTERESTS

Research interest: Hardware Security, Formal Verification, Computer Architecture

EDUCATION

University of North Carolina at Chapel Hill, Chapel Hill, NC, United States — August 2023 — May 2027
Bachelor of Science in Computer Science — Cumulative GPA: 3.94/4.00
Dean's List (2023 Fall - 2026 Spring)

PUBLICATIONS

Published
Jayden Rogers, Niyaz Shakeel, **Xiao Tan**, Samantha Espinosa, Divya Mankani, Cade Chabra, Kaki Ryan, Cynthia Sturton.
Hardware Security Benchmarks for Open-Source SystemVerilog Designs.
In *Proceedings of the IEEE Secure Development Conference (SecDev)*, 2025.

Under Review
Xiao Tan, Cynthia Sturton
CHARGE: Leveraging CWE Hierarchies for Hardware Security SystemVerilog Assertion Generation.

EXPERIENCE

Hardware Security Lab @ UNC — Chapel Hill, NC, United States
Undergraduate Research Assistant — January 2025 — Present

- Developed SystemVerilog Assertions (SVA) to verify security properties of open-source System on Chip (SoC) designs using Cadence JasperGold FPV.
- Maintained and debugged SylQ-SV, a symbolic execution engine for SystemVerilog, by diagnosing execution issues and resolving bugs
- Currently leading a research project on automated hardware security assertion generation for RTL designs.

UNC-Chapel Hill Department of Computer Science — Chapel Hill, NC, United States
Undergraduate Teaching Assistant — June 2025 — December 2025

- Supported 300+ students in COMP 301 (Object-Oriented Programming and Design Patterns) through 10+ weekly office hours across Summer and Fall 2025
- Assisted students in debugging complex Java codebases involving OOP, MVC architecture, and concurrency
- Designed programming assignments and exam review resources as part of the Additional Materials Committee

UNC-Chapel Hill ITS Service Desk — Chapel Hill, NC, United States
Student Assistant - Walk-in Services — June 2025 — December 2025

- Diagnosed and resolved 50+ monthly technical support tickets across hardware, software, and network issues
- Achieved a high first-contact resolution rate by delivering clear, step-by-step technical solutions
- Documented recurring issues to streamline troubleshooting workflows and improve support efficiency

PROJECTS

Verification of Access Control Enforcement in AXI Interconnect — UNC Chapel Hill
COMP 637: Formal Methods for System Security — Spring 2026

- Verified access-control enforcement in an AXI-Lite interconnect extracted from the Hack@DAC21 OpenPiton SoC using assertion-based verification
- Translated 21 access-control property templates from the AKER framework (Restuccia et al., ICCAD 2021)
- Applied Cadence JasperGold FPV and SPV engines to perform model checking and information-flow tracking across controller–peripheral interactions
- Successfully instantiated 17 properties, with 12 satisfied and covered; analyzed 5 violations and identified two inaccuracies in the original AKER templates
- Demonstrated feasibility and limitations of cross-design property translation for hardware security verification

Network Traffic Application Classification via Sequence Models UNC Chapel Hill
COMP 431: Internet Services and Protocols Fall 2025

- Collected real-world network traffic traces using `tcpdump` and constructed a dataset across multiple application types (web, video, gaming, VoIP)
- Extracted structural traffic features including packet sizes, inter-arrival times (IAT), and MTU-based metrics
- Evaluated clustering methods (K-Means, DBSCAN) and observed limited separation across application classes
- Applied sequence models (BiLSTM/LSTM) to capture temporal patterns in traffic flows, achieving ∼96% classification accuracy
- Demonstrated that application types can be inferred from traffic metadata without payload inspection

PRESENTATIONS

Poster: Leveraging CWE Hierarchies for Hardware Security SystemVerilog Assertion Generation
CRA UR2PhD Undergraduate Mentoring Workshop and Research Showcase April 2026
Selected participant; presented research on hardware security and formal verification (travel fully funded)

SELECTED COURSES

Bachelor's Courses

- Formal Methods for System Security
- Digital Logic and Computer Design
- Computer Security Concepts
- Models of Languages and Computation
- *Planned: Operating Systems, Compilers, Machine Learning*

AWARDS

Summer Undergraduate Research Fellowship (SURF) UNC Chapel Hill
Awarded a $4,000 summer research stipend to work on automated hardware security assertion generation. March 2026

SKILLS

- **Languages:** SystemVerilog, Verilog, C/C++, Python, Java, Assembly
- **Formal Verification:** Cadence JasperGold FPV/SPV, SystemVerilog Assertions (SVA)
- **Tools:** Xilinx Vivado, Git, Linux